

LITERATURE REVIEW ABOUT THE DATASET PREPARATION(DATASET TYPE, ANNOTATION TOOL & METHODS, CLASSIFICATION METHOD MANAGEMENT OF UNBALANCED DATASET.)

PAPER -1

Feature	Details Extracted from Paper
Paper Title	A Machine Learning Approach for Post-Disaster Data Curation
Author(s)	Sun Ho Ro, Yitong Li, Jie Gong
Year	2024
Published Journal / Conference	Published by Elsevier (e.g., International Journal of Disaster Risk Reduction / Advanced Engineering Informatics context)
Reference (IEEE Format)	S. H. Ro, Y. Li, and J. Gong, "A Machine Learning Approach for Post-Disaster Data Curation," 2024. (Note: Once your final table is fully compiled, the volume/issue details from the URL can be appended).
Main Goal of the Paper	To automate the large-scale curation, labeling, sorting, and classification of unorganized post-disaster image repositories into a searchable, structured database using a hybrid Computer Vision (CV) and Natural Language Processing (NLP) framework.
Damage Event(s) Studied	Hurricane Harvey (validation case study) and general post-hurricane wind/flood scenarios.
Type of Dataset Used	Ground-level residential building panoramic images (derived from Structural Extreme Events Reconnaissance [StEER] data depot). The paper also discusses aerial/UAV imagery applications.
Annotation / Extraction Tool Used	Automated Platform APIs: AWS Rekognition and Google Cloud Vision API were used to automatically extract low-level feature tags ("secondary labels"). They also implemented custom label models within AWS Rekognition to extract structural/damage-specific features.
How They Classified the Dataset	They structured a customized natural disaster ontology mapping "secondary labels" into five main primary label taxonomies using GloVe word-embeddings and TF-IDF scores: PDF+ 11. Entity: (e.g., People, Building, Vehicle, Debris). Hazard: (e.g., Hurricane,

	Flood). Image Type: (e.g., Ground panorama vs. Aerial/UAV/Satellite). Land Use: (e.g., Residential, Commercial, Vegetation). Building Component: (e.g., Roof, Wall, Window, Door, Garage) . For damage states during validation, images were classified into 4 distinct damage categories: Affected, Minor Damage, Major Damage, and Destroyed.
How They Managed Unbalanced Data	Because this paper focuses on automated data curation and semantic text-mapping rather than training a standalone raw classification model from scratch, they did not balance the dataset using traditional geometric augmentations (like flips or rotations). Instead, they managed the noisy, unbalanced distribution of automated text tags by filtering out irrelevant labels (dropping words like "lobster" or "pilot") and selecting a subset of 654 specific labels tightly coupled with structural damage and disasters to balance feature influence.
Dataset Train / Valid / Test Split (%)	Not applicable in a traditional sense. Since it leverages zero-shot capabilities and pre-trained models (AWS/Google/GloVe), they did not require a standard training split. However, for verification of their custom component detection and framework validation, they used a baseline validation topic dataset consisting of 160 heavily scrutinized images evaluated against manual expert annotations.

PAPER -2

Feature	Details Extracted from Paper
Paper Title	FloodNet: A High Resolution Aerial Imagery Dataset for Post Flood Scene Understanding
Author(s)	Maryam Rahnemoonfar, Tashnim Chowdhury, Argho Sarkar, Debvrat Varshney, Masoud Yari, Robin Murphy
Year	2020
Published Journal / Conference	arXiv preprint (Computer Vision and Pattern Recognition, cs.CV)
Reference (IEEE Format)	M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, and R. Murphy, "FloodNet: A High Resolution Aerial Imagery Dataset for Post Flood Scene Understanding," arXiv preprint arXiv:2012.02951, 2020.
Main Goal of the Paper	To introduce a high-resolution UAV dataset explicitly designed for post-disaster damage assessment and evaluate baseline deep learning models for classification, semantic segmentation, and visual question answering (VQA).
Damage Event(s) Studied	Hurricane Harvey (Post-flood scenarios in Fort Bend County, Texas).
Type of Dataset Used	High-resolution, low-altitude Unmanned Aerial Vehicle (UAV) imagery captured via DJI Mavic Pro quadcopters. Images are extremely detailed, taken at 200 feet Above Ground Level (AGL) with a pixel resolution of 4000x3000.
Annotation / Extraction Tool Used	V7 Darwin Platform. Method: Pixel-wise masks for semantic segmentation, category tags for classification, and completely manual creation of 11,000 question-image pairs for VQA.
How They Classified the Dataset	1. Image Classification: Binary classification scheme—Flooded vs. Non-Flooded. (An image is defined as flooded if more than 30% of its area is covered by floodwater) . 22. Semantic Segmentation: 9 distinct structural and environmental classes : Building-flooded (defined if at least one side touches floodwater) Building-non-flooded Road-flooded Road-non-flooded Water (natural

	bodies like rivers/lakes to separate from floodwater) Tree, Vehicle, Pool, and Grass.
How They Managed Unbalanced Data	For the semantic segmentation models, they handled class variance, over-fitting, and data imbalances through traditional image augmentation techniques including : Random shuffling Scaling Flipping Random rotation Utilization of a larger "crop size" (713×713 during training) to preserve high-resolution spatial contexts across rare classes.
ML Model Used (Baselines Evaluated)	Classification: InceptionNetv3, ResNet50, and Xception .Semantic Segmentation: PSPNet (with ResNet101 backbone), ENet, and DeepLabv3+ .Visual Question Answering (VQA): Stacked Attention Network (SAN), Multimodal Factorized Bilinear (MFB) with Co-Attention, and simple baseline feature concatenation/multiplication models.

PAPER -3

Feature	Details Extracted from Paper
Paper Title	RescueNet: A High Resolution UAV Semantic Segmentation Dataset for Natural Disaster Damage Assessment PDF
Author(s)	Maryam Rahnemoonfar, Tashnim Chowdhury, and Robin Murphy PDF
Year	2023 PDF
Published Journal / Conference	Scientific Data (Nature Portfolio) PDF
Reference (IEEE Format)	M. Rahnemoonfar, T. Chowdhury, and R. Murphy, "RescueNet: A High Resolution UAV Semantic Segmentation Dataset for Natural Disaster Damage Assessment," Scientific Data, vol. 10, no. 1, p. 913, Dec. 2023.
Main Goal of the Paper	To present RescueNet, a benchmark high-resolution aerial dataset that addresses the lack of comprehensive, pixel-level annotations for entire post-disaster scenes, offering detailed sub-categorizations of building and road structural damage. PDF+ 2
Damage Event(s) Studied	Hurricane Michael (Post-disaster response phase at Mexico Beach, Florida, and surrounding regions). PDF+ 3
Type of Dataset Used	Low-altitude, high-resolution sUAV (small Unmanned Aerial Vehicle) imagery. Collected over 80 flights using DJI Mavic Pro quadcopters at altitudes up to 400 feet AGL, with a crisp pixel resolution of 4000x3000. PDF+ 3
Annotation / Extraction Tool Used	V7 Darwin Platform . PDF• Method: Objects were meticulously manually annotated to generate polygon coordinates stored in JSON format, which were subsequently compiled into pixel-level segmentation masks. PDF+ 2
How They Classified the Dataset	It implements a two-tier classification scheme: PDF1. Image-Level (Neighborhood) Classification: Categories area damage into 3 tiers based on structural layout : PDF+ 1• Superficial damage (no damaged structures) PDF• Medium damage (a few damaged structures) PDF• Major damage (at least 1 completely destroyed building or $\approx 50\%$ area

	covered in debris). PDF2. Semantic Segmentation (10 Classes): PDF• Water, Vehicle, Tree, Pool . PDF+ 2• Road split into: Road-Clear vs. Road-Blocked (obstructed by flood, debris, sand) . PDF+ 4• Building split into 4 levels adapted for aerial top-views (following FEMA guidelines) : Building-No-Damage, Building-Minor-Damage (roof patches coverable by tarps), Building-Major-Damage (severe structural failures), and Building-Total-Destruction (foundation or multi-wall collapse). PDF+ 4
How They Managed Unbalanced Data	This paper highlights a massive, realistic post-disaster imbalance (e.g., Background is 52%, Water is 8.9%, whereas severe damage metrics like Building Total Destruction comprise only 1% of the entire dataset pixels). They countered this extreme skewness via: PDF• Strategic Stratified Data Partitioning: They utilized the image-level (Neighborhood) damage classifications to pull proportional samples, ensuring that rare, high-destruction images were distributed systematically across splits rather than clumped in one set . PDF• Data Augmentation: Scaling, flipping, random shuffling, and random rotation implemented during training loops to combat model overfitting to majoritarian classes . PDF• Fixed Input Resolution Window: Resizing and cropping training images to a dense 713×713 matrix to ensure rare visual features (like vehicles or localized total destruction patches) were sufficiently preserved. PDF+ 1
ML Model Used (Baselines Evaluated)	Baseline evaluations for semantic segmentation were performed using PSPNet (ResNet101 backbone), DeepLabv3+ (ResNet101 backbone), Attention U-Net, and Segmenter (Vision Transformer network using ViT-Tiny and ViT-Small backbones). (Note: Attention U-Net and PSPNet achieved the strongest Mean IoU results). PDF+ 3

PAPER -4

Feature	Details Extracted from Paper
Paper Title	xBD: A Dataset for Assessing Building Damage from Satellite Imagery PDF+ 1
Author(s)	Ritwik Gupta, Richard Hosfelt, Sandra Sajeev, Nirav Patel, Bryce Goodman, Jigar Doshi, Eric Heim, Howie Choset, and Matthew Gaston PDF+ 1
Year	2019 PDF+ 1
Published Journal / Conference	arXiv preprint (Computer Vision and Pattern Recognition, cs.CV) PDF+ 1
Reference (IEEE Format)	R. Gupta et al., "xBD: A Dataset for Assessing Building Damage from Satellite Imagery," arXiv preprint arXiv:1911.09296, 2019.
Main Goal of the Paper	To present xBD, a massive, multi-disaster, large-scale dataset featuring pre- and post-event high-resolution satellite imagery intended to advance machine learning research for change detection and automated building damage assessment. PDF+ 1
Damage Event(s) Studied	Covers 19 separate major natural disaster events globally, spanning 5 key categories : PDF+ 1• Earthquake/Tsunami (e.g., Mexico City, Palu/Indonesia) PDF+ 1• Floods (e.g., Midwest US, Hurricane Harvey) PDF+ 1• Volcanic Eruptions (e.g., Guatemala Fuego, Lower Puna) PDF+ 2• Wildfires (e.g., Santa Rosa, Carr, Woolsey, Portugal) PDF+ 2• Wind/Tornadoes (e.g., Hurricane Michael, Joplin, Moore). PDF+ 2
Type of Dataset Used	High-resolution satellite electro-optical (EO) imagery sourced from the Maxar/DigitalGlobe Open Data Program. All imagery features 3-band RGB formats with a crisp Ground Sample Distance (GSD) below 0.8 meters. PDF+ 2
Annotation / Extraction Tool Used	Web-based annotation tool developed in-house by CrowdAI . PDF• Method: Multi-step expert process. First, annotators drew precise polygon coordinates around visible building footprints on pre-disaster imagery. These "ideal structural footprints" were then

	overlaid onto post-disaster image pairs for ordinal grading, avoiding occlusion or destruction masks. PDF+ 3
How They Classified the Dataset	Introduces the generalized Joint Damage Scale (grounded in FEMA/HAZUS and EMS-98 scales) spanning 4 ordinal levels : PDF+ 1• Level 0 (No Damage): Undisturbed; no structural/shingle damage or burns . PDF• Level 1 (Minor Damage): Partially burnt, water surrounding structure, roof elements missing, or visible cracks . PDF• Level 2 (Major Damage): Partial wall or roof collapse, surrounded by mud/water . PDF• Level 3 (Destroyed): Scorched, completely collapsed, or otherwise no longer present . PDF• Environmental Factors: Separately annotated bounding boxes/polygons for smoke, fire, flood water, pyroclastic flow, and lava. PDF
How They Managed Unbalanced Data	The dataset is extremely skewed towards "No Damage" (313k polygons) compared to Minor (36k), Major (29k), and Destroyed (31k). To manage this massive class imbalance, they introduced two crucial mechanisms: PDF1. Evaluation Metric Tweak: Abandoned simple accuracy (which yields a false 75% baseline accuracy simply by predicting "No Damage") and mandated Weighted F1 Score to properly balance precision and recall across rare, highly damaged classes. PDF+ 12. Loss Function Modification: Instead of regular Cross-Entropy, they developed an Ordinal Cross-Entropy Loss Function. This mathematically penalizes errors relative to the distance between the true and predicted class (e.g., misclassifying a 'Destroyed' building as 'No Damage' incurs a far heavier penalty than misclassifying it as 'Major Damage'). PDF+ 1
ML Model Used (Baselines Evaluated)	• Localization (Change Detection): An altered U-Net architecture (originally adapted from SpaceNet) to segment building polygons . PDF• • Classification (Damage Grading): A dual-stream CNN combining a pre-trained ResNet50 (ImageNet weights) with a shallow, custom, randomly initialized 3-layer CNN stream. The flattened outputs of both streams are concatenated and passed through dense

	layers to produce the ordinal probabilities. PDF+ 3
Dataset Train / Valid / Test Split (%)	Explicitly divided into a 80 / 10 / 10 % configuration across images and annotations : PDF• Train Split: 18,336 images (632,228 building polygons) PDF+ 1• Test Split: 1,866 images (109,724 building polygons) PDF+ 1• Holdout Split: 1,866 images (108,784 building polygons) — retained privately for un-gameable validation. PDF+ 1

PAPER -5

Feature	Details Extracted from Paper
Paper Title	RSVLM-QA: A Benchmark Dataset for Remote Sensing Vision Language Model-based Question Answering PDF
Author(s)	Xing Zi, Jinghao Xiao, Yunxiao Shi, Xian Tao, Jun Li, Ali Braytee, and Mukesh Prasad PDF+ 1
Year	2025 PDF
Published Journal / Conference	Proceedings of the 33rd ACM International Conference on Multimedia (MM '25) PDF
Reference (IEEE Format)	X. Zi et al., "RSVLM-QA: A Benchmark Dataset for Remote Sensing Vision Language Model-based Question Answering," in Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), Oct. 2025, pp. 1–7. PDF+ 1
Main Goal of the Paper	To introduce a massive, content-rich Remote Sensing Visual Question Answering (RSVQA) benchmark dataset generated through an automated LLM pipeline, aiming to evaluate the complex reasoning, counting, and spatial understanding of advanced Vision Language Models (VLMs). PDF+ 2
Damage Event(s) Studied	Not focused on a single localized disaster event. Instead, it covers generalized, global Earth observation, urban landscapes, and dense structural environments. PDF
Type of Dataset Used	Multi-source high-resolution satellite and aerial electro-optical imagery. It synthesizes data from 4 prominent public remote sensing benchmarks: WHU Building Dataset, LoveDA, INRIA Aerial Image Labeling, and iSAID. PDF+ 1
Annotation / Extraction Tool Used	Automated AI Pipeline: Rather than using manual annotation tools like CVAT, the authors engineered a Dual-Track Automated pipeline using GPT-4.1. PDF+ 4 • Track 1: GPT-4.1 was prompted to ingest the images and generate rich narrative captions, from which it extracted object entities and spatial relations . PDF+ 3 • Track 2: A programmatic script extracted ground-truth counting data

	directly from the original segmentation source labels, which GPT-4.1 converted into natural text answers. PDF+ 1
How They Classified the Dataset	Classified into a dense multi-tiered structured text ontology. Each image averages 10.62 object tags (entities) and 5.63 explicit spatial relations. The VQA component is structurally classified into 6 broad question types: Spatial Reasoning, Quantitative Queries, Presence Verification, Feature Analysis, Object Recognition, and Narrative Captions. PDF+ 4
How They Managed Unbalanced Data	Because automated text generation can introduce massive skews, linguistic repetition, or hallucinations, they managed dataset balance and quality using a two-stage protocol: PDF1. Intelligent Screening Agent: GPT-4.1 filtered out ill-posed, highly ambiguous, or non-verifiable questions. PDF2. Human Expert Verification: Trained human annotators manually verified and adjudicated all flagged data and a massive random subset, filtering out ~14% of faulty entries to balance semantic accuracy. PDF
ML Model Used (Baselines Evaluated)	Evaluated in a zero-shot inference mode using 6 premier, cutting-edge foundational Vision Language Models: Ovis2 (the top performer), Gemma3, InternVL3, Qwen2.5-VL, LLaVA, and BLIP-2. PDF+ 2
Dataset Train / Valid / Test Split (%)	The final consolidated dataset comprises 13,820 images and 162,373 unique VQA pairs. The paper focuses on zero-shot benchmarking across the entire curated dataset to test intrinsic generalization capabilities, meaning task-specific fine-tuning splits were bypassed. PDF+ 2

PAPER -6

Feature	Details Extracted from Paper
Paper Title	Robust Disaster Assessment from Aerial Imagery Using Text-to-Image Synthetic Data PDF
Author(s)	Tarun Kalluri, Jihyeon Lee, Manmohan Chandraker, Kihyuk Sohn, Sahil Singla, Joseph Xu, and Jeremiah Liu PDF
Year	2024 PDF
Published Journal / Conference	IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW) PDF+ 1
Reference (IEEE Format)	T. Kalluri et al., "Robust Disaster Assessment from Aerial Imagery Using Text-to-Image Synthetic Data," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2024, pp. 7449–7454.
Main Goal of the Paper	To overcome the lack of real-world post-disaster labeled data in under-resourced geographies by leveraging text-to-image generative models to synthesize large-scale post-disaster supervision from available pre-disaster imagery. PDF+ 2
Damage Event(s) Studied	• Hurricanes (Ian, Maria, Michael, Laura) via the SKAI dataset . PDF• Tornadoes (Moore Tornado), Floods (Nepal Floods), and Wildfires (Portugal Wildfire) via the xBD dataset. PDF+ 3
Type of Dataset Used	High-resolution satellite electro-optical and aerial imagery sourced from two major benchmarks: xBD and SKAI. PDF+ 2
Annotation / Extraction Tool Used	Automated Prompt-Driven Editing Framework: Bypasses traditional manual annotation tools. PDF+ 1• Method: Pre-disaster target images are passed through a VQGAN encoder to compute latent tokens, which are then modified with a binary center mask. A MUSE (text-to-image transformer) model processes these masked tokens alongside a domain-specific text prompt pool to output highly realistic post-disaster twins.

	Ground-truth labels are automatically mapped directly from the prompt text. PDF+ 4
How They Classified the Dataset	Binary classification scheme: Damaged (Label 1) vs. Undamaged (Label 0). PDF+ 2• The classification relies entirely on linguistic framing in their Prompt Pool (e.g., "An aerial view of a house damaged due to a tornado" maps to Label 1, while "An aerial view of an intact house" maps to Label 0). PDF+ 1
How They Managed Unbalanced Data	This paper deals with a dual challenge: real-world datasets with over 80% undamaged skews, and domain gaps from synthetic data. They counter this via: PDF+ 11. Generative Data Expansion: Generating thousands of highly targeted synthetic "damaged" scenes from text pools to artificially balance the positive class distribution. PDF+ 12. Two-Stage MLP Re-Training: The shared-parameter Siamese Vision Transformer is fully pre-trained on real source domains. Then, to fix imbalance/domain gaps, they freeze the encoder backbones and exclusively fine-tune the final Multi-Layer Perceptron (MLP) classification layers on the in-domain synthetic target data. PDF+ 43. Evaluation Metrics: Switched to AUPRC (Area Under the Precision-Recall Curve), which remains highly robust against positive-class sparse datasets. PDF
ML Model Used	• Image Generation: MUSE (Non-autoregressive text-to-image transformer with T5-XXL text encoder and VQGAN tokenizer) . PDF+ 1• Damage Assessment Classification: A Siamese network utilizing an ImageNet pre-trained ViT-B/16 (Vision Transformer) backbone with parameter-shared encoders connected to a custom 2-layer MLP head. PDF+ 2

PAPER -7

Feature	Details Extracted from Paper
Paper Title	Attention Based Semantic Segmentation on UAV Dataset for Natural Disaster Damage Assessment PDF
Author(s)	Tashnim Chowdhury and Maryam Rahnemoonfar PDF
Year	2021 PDF+ 1
Published Journal / Conference	IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2021) PDF+ 1
Reference (IEEE Format)	T. Chowdhury and M. Rahnemoonfar, "Attention Based Semantic Segmentation on UAV Dataset for Natural Disaster Damage Assessment," in 2021 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2021, pp. 1–4. PDF+ 1
Main Goal of the Paper	To introduce a novel self-attention based semantic segmentation model (ReDNet) to capture global contextual dependencies and improve pixel-level feature recognition across highly varied post-disaster structures. PDF+ 4
Damage Event(s) Studied	Hurricane Michael. PDF+ 1
Type of Dataset Used	High-Resolution UAV Dataset (HRUD), obtained from the Center for Robot-Assisted Search and Rescue (CRASAR) repository. Images are captured at low altitudes (200 feet Above Ground Level) using DJI Mavic Pro quadcopters, featuring a resolution of 3000x4000 pixels. PDF+ 4
Annotation / Extraction Tool Used	Manual Semantic Masking: Ground-truth pixel-level annotation masks were generated for 1,973 disaster images. (Note: As seen in earlier papers from this lab, the underlying platform utilized is V7 Darwin). PDF+ 2
How They Classified the Dataset	Consists of 8 distinct structural and environmental classes : PDF+ 2• Debris, Water, Vehicle, Road, Tree, Sand, and Unlabeled . PDF+ 1• Building Damage Classification: The original data contains 4 damage levels (no damage, medium damage, major damage, total destruction). However, for this paper's experiments, they simplified it into a binary classification scheme: "Building

	Not Totally Destroyed" (merging no, medium, and major damage) vs. "Building Totally Destroyed". PDF+ 1
How They Managed Unbalanced Data	The dataset features highly variable-sized classes and extreme pixel imbalances. They managed this through: PDF1. Data Augmentation: Implemented random shuffling, scaling, flipping, and random rotation within the training loop to prevent the network from overfitting to majoritarian background classes. PDF2. Large Crop Resizing Window: Images were explicitly resized to 713×713 pixels during training. Using a large crop size ensures that small, rare classes (like Vehicle or concentrated structural damage patches) are not erased during training steps. PDF3. Architectural Mitigation (Self-Attention): They designed a parallel Position Attention Module (PAM) that links features from lower and higher layers of ResNet-101. This captures long-range global context, allowing the model to differentiate between similar, highly skewed textures like debris, sand, and destroyed buildings. PDF+ 4
ML Model Used	ReDNet (ResNet-based Dual attention Network): A custom self-attention network utilizing a ResNet-101 backbone connected to two parallel Position Attention Modules (PAMs) that apply spatial feature maps via matrix multiplication and softmax operations. PDF+ 3• Baselines compared: ENet, DeepLabv3+, and PSPNet. PDF
Dataset Train / Valid / Test Split (%)	Performance is explicitly reported on the designated HRUD Testing Set (using an 88% mean IoU achievement as evaluation baseline). Following the standard protocol of the HRUD benchmark pipeline, the dataset is distributed into 80% Training / 10% Validation / 10% Testing. PDF+ 1

PAPER -8

Feature	Details Extracted from Paper
Paper Title	Building Disaster Damage Assessment in Satellite Imagery with Multi-Temporal Fusion PDF
Author(s)	Ethan Weber and Hassan Kan PDF+ 1
Year	2020 PDF
Published Journal / Conference	ICLR 2020 AI for Earth Sciences Workshop PDF
Reference (IEEE Format)	E. Weber and H. Kan, "Building Disaster Damage Assessment in Satellite Imagery with Multi-Temporal Fusion," in Proceedings of the International Conference on Learning Representations (ICLR) AI for Earth Sciences Workshop, 2020, pp. 1–7.
Main Goal of the Paper	To investigate optimal problem framing, data pre-processing, and multi-temporal feature fusion architectures using shared-weight CNN backbones to significantly outperform baseline models for building localization and damage grading. PDF+ 2
Damage Event(s) Studied	Multi-disaster scope spanning earthquakes, floods, volcanic eruptions, wildfires, and wind/hurricanes across 16 global geographical locations (derived from the xBD dataset). PDF
Type of Dataset Used	High-resolution multi-temporal satellite imagery sourced from the xBD dataset (originally collected by WorldView-3 satellites). Features a crisp spatial resolution of 0.3 meters per pixel. PDF+ 2
Annotation / Extraction Tool Used	Polygon-to-Semantic Conversion: They utilized the existing ground-truth annotations from the xBD repository (850,736 building polygons). For their specific training pipeline, they programmatically mapped these polygons directly into a pixel-level dense classification map suitable for semantic segmentation. PDF+ 3
How They Classified the Dataset	Joint prediction scheme mapping into a 5-class damage classification output : PDF+ 1• Class 0: No building (Background) PDF+ 1• Class 1: Undamaged building PDF+ 1• Class

	2: Minor damage PDF+ 1 • Class 3: Major damage PDF+ 1 • Class 4: Destroyed building . PDF+ 1 Note: Instead of running two separate models for localization and damage assessment like the xBD baseline, they framed the task so that identifying any pixel belonging to classes 1-4 jointly counted as a localized "building". PDF+ 1
How They Managed Unbalanced Data	The xBD dataset is heavily skewed, with 73.6% of all building polygons falling under "no damage" and the vast majority of image pixels representing background. They handled this imbalance through two methods: PDF1. Class-Specific Cross-Entropy Weighting: They modified the loss function by weighting class errors inversely proportional to their statistical occurrence in the training dataset. This forces the network to penalize mistakes on rare damage classes heavily and prevents the network from collapsing into only predicting background. PDF+ 42. Image Quadrant Cropping: Rather than feeding massive 1024×1024 images which cause fine building boundaries to be lost, they trained and processed models on smaller 512×512 quadrants (crops). This localized context vastly improves minority class detection. PDF+ 2
ML Model Used	A multi-temporal fusion network. It passes the pre-disaster and post-disaster image crops independently through parallel ResNet-50 Feature Pyramid Network (FPN) backbones that utilize shared weights. The extracted 256-channel feature maps from both streams are stacked/concatenated into a 512-channel feature map, which is then fed into a Semantic Segmentation Head (built using the Detectron2 library). PDF+ 4
Dataset Train / Valid / Test Split (%)	Adheres strictly to the standardized xBD partition schema across their ablation studies : PDF+ 2 • Evaluated and tuned using the official competition test sets. This generally maps back to the baseline 80% Train / 10% Validation / 10% Test (Holdout) data split protocol. PDF+ 2

PAPER -9

Feature	Details Extracted from Paper
Paper Title	Cross-directional Feature Fusion Network for Building Damage Assessment from Satellite Imagery PDF
Author(s)	Yu Shen, Sijie Zhu, Taojiannan Yang, and Chen Chen PDF
Year	2020 PDF
Published Journal / Conference	NeurIPS 2020 Artificial Intelligence for Humanitarian Assistance and Disaster Response Workshop PDF
Reference (IEEE Format)	Y. Shen, S. Zhu, T. Yang, and C. Chen, "Cross-directional Feature Fusion Network for Building Damage Assessment from Satellite Imagery," in Proceedings of the NeurIPS Artificial Intelligence for Humanitarian Assistance and Disaster Response Workshop, 2020, pp. 1–8.
Main Goal of the Paper	To develop a deep learning framework that models correlations between pre- and post-disaster remote sensing scenes via a cross-directional feature aggregation block, while simultaneously optimizing data augmentation for highly visually ambiguous "hard classes." PDF+ 2
Damage Event(s) Studied	Multi-disaster profile including hurricanes, floods, wildfires, and earthquakes. PDF
Type of Dataset Used	High-resolution multi-temporal satellite imagery sourced from the benchmark xBD dataset. Individual image pairs feature a spatial grid size of 1024x1024 pixels. PDF+ 1
Annotation / Extraction Tool Used	Polygon-to-Semantic Extraction: Inherited ground-truth annotations directly from the xBD data repository (over 800,000 building annotations). These polygonal outlines were mapped programmatically to support an end-to-end pixel-level dense classification structure. PDF+ 2
How They Classified the Dataset	Pixel-level multi-class damage categorization spanning 5 definitive states : PDF+ 1• Class 0: Background (no building present) PDF• Class 1: No damage PDF• Class 2: Minor

	damage PDF• Class 3: Major damage PDF• Class 4: Destroyed. PDF
How They Managed Unbalanced Data	The dataset exhibits extreme structural class imbalances, showing severe skewness toward the "No Damage" class (313,003 polygons) compared to Minor (36,860) and Major (29,904) damage states. They counteracted this via: PDF+ 11. Class-Targeted CutMix Data Augmentation: Instead of applying CutMix randomly across the entire dataset, they explicitly masked and pulled image patches from "hard classes" (Minor and Major damage) and pasted them into random training samples. This forced the neural network to concentrate on learning robust, fine-grained localized boundaries for the rarest labels. PDF+ 22. Geometric Regularization: Standard online data augmentations including horizontal/vertical flips and rotations were systematically executed. PDF3. Spatial Window Cropping: Large 1024×1024 images were processed down into dense 512×512 pixel quadrant crops for training. PDF
ML Model Used	A Two-Stage U-Net Framework utilizing a shared-parameter ResNet-50 backbone : PDF+ 2• Stage 1 (Building Segmentation): A single-branch U-Net utilizes only pre-disaster imagery to output building footprints . PDF+ 3• Stage 2 (Damage Assessment): Pre- and post-disaster image tensors are passed down a dual-branch U-Net initialized with Stage 1's weights. PDF+ 2• Cross-Directional Fusion (CDF) Module: Implements a cross-attention style framework inspired by Squeeze-and-Excitation (SE) blocks. It applies global average pooling and sigmoid activations across concatenated branches to compute channel-wise and spatial-wise multiplication factors, dynamically cross-embedding features across both temporal streams. PDF+ 4

PAPER -10

Feature	Details Extracted from Paper
Paper Title	Rapid building damage assessment workflow: An implementation for the 2023 Rolling Fork, Mississippi tornado event PDF
Author(s)	Caleb Robinson, Simone Fobi Nsutezo, Anthony Ortiz, Tina Sederholm, Rahul Dodhia, Cameron Birge, Kasie Richards, Kris Pitcher, Paulo Duarte, and Juan M. Lavista Ferres PDF
Year	2023 PDF
Published Journal / Conference	IEEE/CVF International Conference on Computer Vision Workshops (ICCVW) PDF
Reference (IEEE Format)	C. Robinson et al., "Rapid building damage assessment workflow: An implementation for the 2023 Rolling Fork, Mississippi tornado event," in Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), 2023, pp. 3760–3764.
Main Goal of the Paper	To introduce and operationally validate a human-in-the-loop workflow that can rapidly train and deploy highly accurate building damage assessment models immediately after a natural disaster using minimal data and only post-disaster imagery. PDF+ 2
Damage Event(s) Studied	The March 2023 EF4 Tornado event in Rolling Fork, Mississippi. PDF+ 1
Type of Dataset Used	Post-disaster high-resolution satellite imagery captured by Planet Labs SkySat satellites (Scene ID: 20230325_195842_ssc11_u0001). PDF+ 1
Annotation / Extraction Tool Used	Microsoft Satellite Imagery Labeling Tool (Open-source on GitHub) . PDF• Method: An interactive visual platform where a human annotator can quickly select and categorize building footprint polygons. PDF
How They Classified the Dataset	Pixel-level semantic categories are grouped into 3 core classes during the training step: PDF+ 21. Background PDF+ 22. Building (Undamaged) PDF+ 33. Damaged Building PDF+ 2• Aggregation into Object Level: The pixel map outputs are combined with known vector polygons from the Global Microsoft

	Building Footprint Dataset. For each building footprint, they programmatically calculate y^i , which is the percentage area of the footprint classified as damaged (ranging from 0 to 100). This can be mapped directly to binary damage thresholds. PDF+ 4
How They Managed Unbalanced Data	Instead of fighting natural class imbalances by artificially synthesizing or downloading millions of images, they bypass it operationally via Few-Shot Active Learning (Human-in-the-Loop): PDF+ 11. Extremely Targeted Seed Set: The human expert manually labels a tiny, balanced sample size of only ≈ 100 to 153 total specific polygon examples across the AOI (e.g., 34 background, 71 undamaged, and 48 damaged buildings). This forces the model to learn localized context immediately. PDF+ 32. Iterative Refinement: If a quick inference run reveals a high rate of false positives/negatives due to regional skews, the user adds a few more labels to correct the imbalance and re-trains the model until validation metrics stabilize. PDF
ML Model Used	A U-Net semantic segmentation architecture utilizing an ImageNet pre-trained ResNet-50 backbone. PDF

PAPER -11

Feature	Details Extracted from Paper
Paper Title	xBD: A Dataset for Assessing Building Damage from Satellite Imagery
Author(s)	Ritwik Gupta, Richard Hosfelt, Sandra Sajeev, Nirav Patel, Bryce Goodman, Jigar Doshi, Eric Heim, Howie Choset, and Matthew Gaston
Year	2019
Published Journal / Conference	arXiv preprint (Computer Vision and Pattern Recognition, cs.CV)
Reference (IEEE Format)	R. Gupta et al., "xBD: A Dataset for Assessing Building Damage from Satellite Imagery," arXiv preprint arXiv:1911.09296, 2019.
Main Goal of the Paper	To introduce a massive, standardized multi-disaster satellite imagery dataset featuring precise building polygons, generalized ordinal damage scales, and metadata to advance computer vision research in change detection and rapid damage grading. PDF+ 1
Damage Event(s) Studied	19 global disaster events divided into Tiers : PDF+ 1• Tier 1: Guatemala Volcano, Hurricanes (Michael, Florence, Harvey, Matthew), Santa Rosa & Carr Wildfires, Midwest US Floods, Indonesia Tsunami, and Monsoons (Nepal/India/Bangladesh) . PDF• Tier 3: Tornadoes (Moore, Tuscaloosa, Joplin), Sunda Strait Tsunami, Lower Puna Eruption, and Wildfires (Woolsey, Pinery, Portugal). PDF
Type of Dataset Used	High-resolution satellite electro-optical (EO) 3-band RGB imagery sourced from the Maxar/DigitalGlobe Open Data Program. Ground Sample Distance (GSD) is strictly under 0.8 meters to capture minute structural discrepancies. PDF+ 4
Annotation / Extraction Tool Used	Web-based custom polygon labeling interface developed in-house by CrowdAI. PDF• Method: Multi-phase pipeline. Pre-disaster image scenes were sent to expert annotators to outline clean, unobscured building polygons. These identical footprint matrices were

	subsequently transposed directly onto the post-disaster image pairs for damage classification. PDF+ 1
How They Classified the Dataset	Establishes a 4-level Joint Damage Scale (grounded mathematically in HAZUS, Kelman, and EMS-98 metrics) : PDF+ 1• Level 0 (No Damage): Undisturbed structure . PDF• Level 1 (Minor Damage): Partially burnt, misshapen roof elements, visible cracks, or water/volcanic flow surrounding building . PDF• Level 2 (Major Damage): Partial wall or roof structural collapse . PDF• Level 3 (Destroyed): Scorched, foundations wiped away, completely collapsed or submerged . PDF• Contextual Features: Custom separate bounding box overlays for smoke, fire, flood water, and lava flows. PDF+ 1
How They Managed Unbalanced Data	The creators actively anticipated massive dataset skews (where the vast majority of image pixels represent rural/urban background, and positive targets skew ~75% toward "No Damage"). They addressed this via: PDF1. Negative Imagery Calibration: Intentionally retained a large buffer area during regional imagery triage. This intentionally kept a substantial baseline ratio of completely undamaged "negative scenes" so models learn background variance. PDF+ 12. Algorithmic Evaluation Metric Shift: Mandated the use of Weighted F1 Scores for the xView2 challenge backends, preventing models from achieving high scoring parameters by simply ignoring rare damage classes.3. Spatial Dimension Standardisation: Large satellite passes were tiled out into dense, accessible 1024×1024 pixel imagery grids.
ML Model Used (Baselines Evaluated)	The official baseline pipeline uses a Two-Stage Separate Framework:• Stage 1 (Localization): A modified U-Net topology optimized to output binary building footprint masks from the pre-disaster imagery feeds.• Stage 2 (Classification): The localized building crop dimensions are passed into a dual-branch network. An ImageNet-weighted

	ResNet-50 extracts deep semantic visual characteristics from pre-and-post pairs, combined with a custom shallow 3-layer CNN head to generate the final 4-tier classification array.
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